

research : API, REST API, Types (GET, POST, PUT, DELETE,...)

API and REST API

An API (Application Programming Interface) is a set of rules and protocols that allows different software applications to communicate with each other. APIs help developers connect systems, share data, and use services without needing to understand the internal code of another application. For example, when a mobile application shows weather information or allows users to log in using Google or Facebook, it uses APIs to exchange data with external servers.

One of the most popular types of APIs is the REST API. REST stands for Representational State Transfer. It is an architectural style that uses the HTTP protocol to send and receive data between a client and a server. REST APIs are widely used in web and mobile applications because they are simple, fast, and flexible. Most REST APIs exchange data in JSON format, which is easy for both humans and computers to read. REST APIs work through different HTTP request methods. Each method has a specific purpose:

1. GET

The **GET** method is used to retrieve data from a server without changing it.

2. POST

The **POST** method is used to create or add new data to the server.

3. PUT

The **PUT** method is used to update existing data completely.

4. DELETE

The **DELETE** method is used to remove data from the server.

5. PATCH

The **PATCH** method is used to update only part of the data instead of replacing all of it.

